PARISHRAM 2.0

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max : 10:00

CLASS 12TH 2026

Electric Charges and Fields

PHYSICS LECTURE-2

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TOPICS TO BE COVERED

1. Properties of electric charge
2. Coulomb’s law
3. Vector form of coulomb’s law
   principle of superposition, forces between multiple charges, Electric field, Electric field due to a point charge

My Parishram 2.0 students during their first physics class, when I invested first 30 minutes in conversation and motivation





Recap :-

- Electric Charge
- Charging methods
  - Friction
  - Induction
  - Conduction.
- Conductors & Insulators.

Unit of Charge

Charge → mass

माल भरा हुआ है!

→ S.I. unit ⇒ **Coulomb (C)**

→ C.G.S unit → ESU ⟹ $1\ ESU = \frac{1}{3\times10^9}$ Coulomb

Electrostatic Unit of charge

→ EMU → 1 EMU = 10 Coulombs.

Electromagnetic unit of charge.

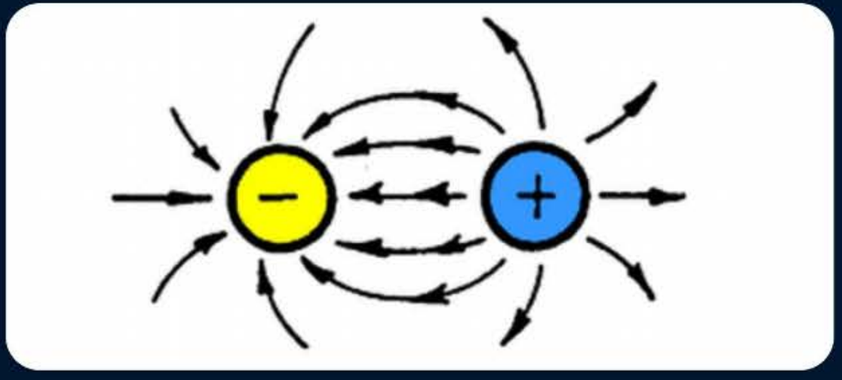
Intrinsic property of matter.

Properties of Electric Charge

It also has properties.

- Charge is invariant
- Accelerated charge radiates energy → Atoms | Nuclei.

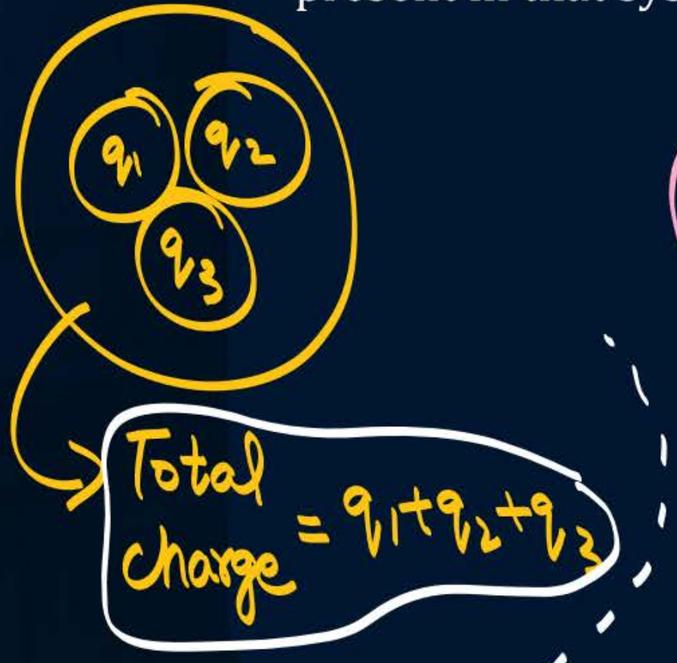
- Additivity of Charges
- Conservation of Charge
- Quantization of Charge

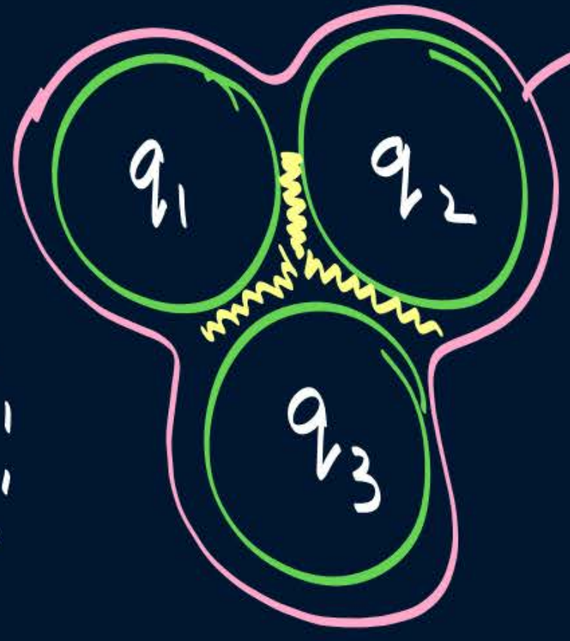


Additivity of Charges

$|e^-| = |p^+| = 1.6 \times 10^{-19}\,C$

- The **total electric charge** of a system is the **algebraic sum** of all the individual charges present in that system.





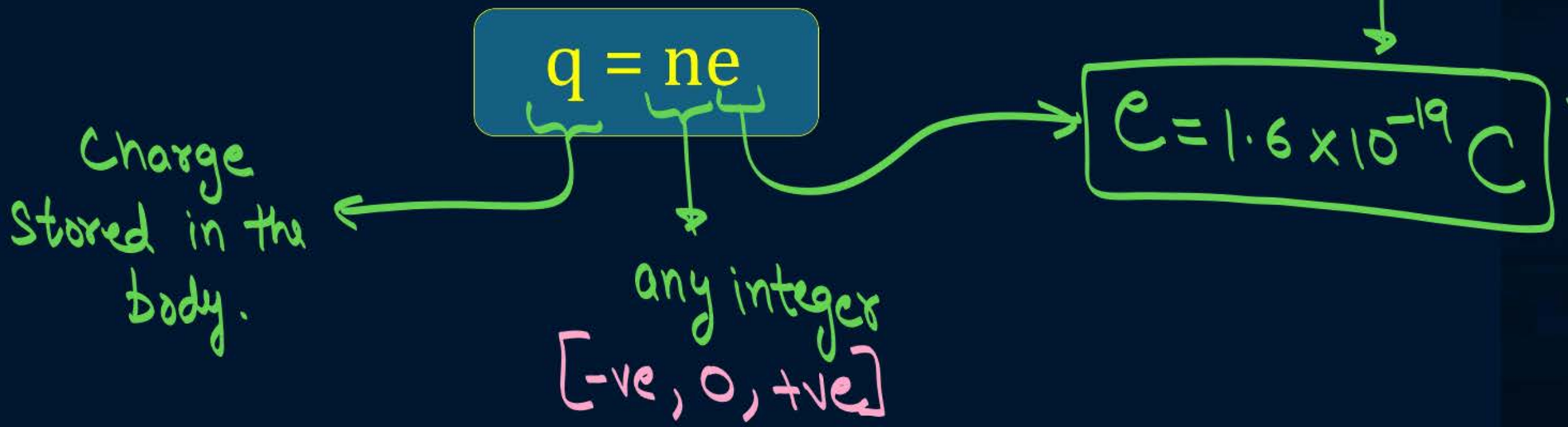
Total Charge in the Newly formed body :-

Total charge $= q_1+q_2+q_3$

Quantization of Charge

- Every free charge is an integral multiple of a fundamental charge unit denoted by e. Thus charge q of a body is always given by

$$q = ne$$



Conservation of Charge

- **Law of Conservation of Charge:**

Charge on a body can neither be created, nor be destroyed, it can only be transferred from one body to another.

This is valid for an isolated system.

Initial Charge on the body = Final charge on the body

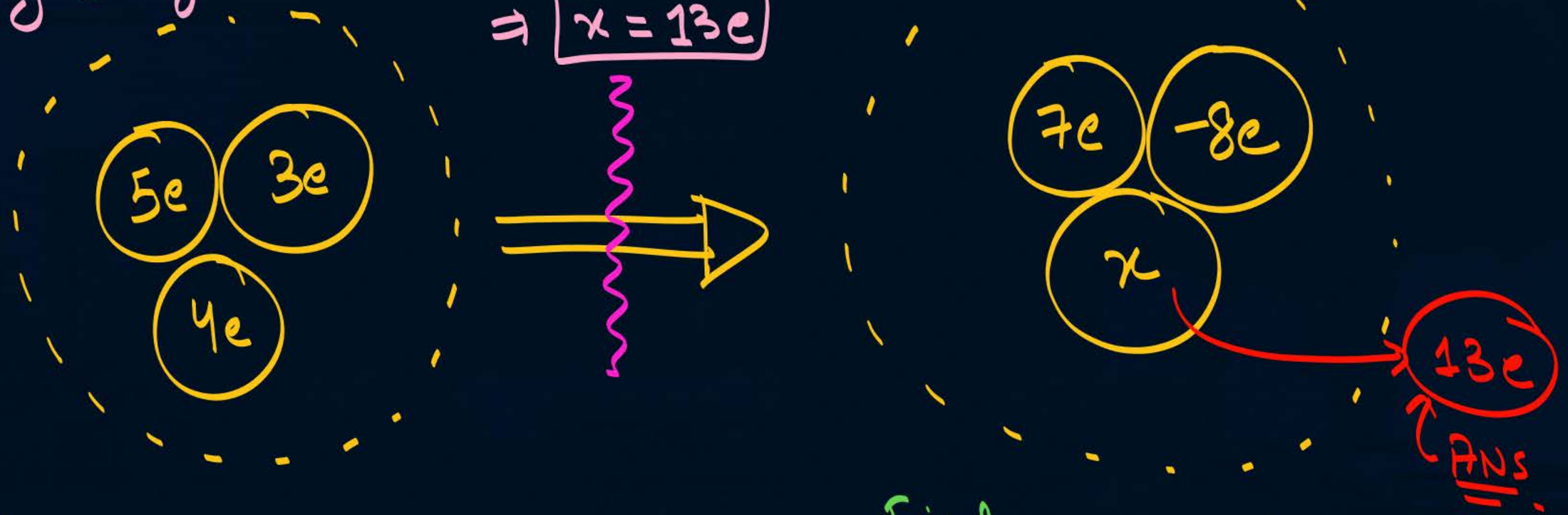
By law of Conservation of charge.

Initial charge = Final charge

$\Rightarrow 12e = -1e + x$

$\Rightarrow \boxed{x = 13e}$

ex:



Initial charge of the system $= 5e + 3e + 4e$

$= \boxed{12e}$

Final charge of the system $= 7e + (-8e) + x$

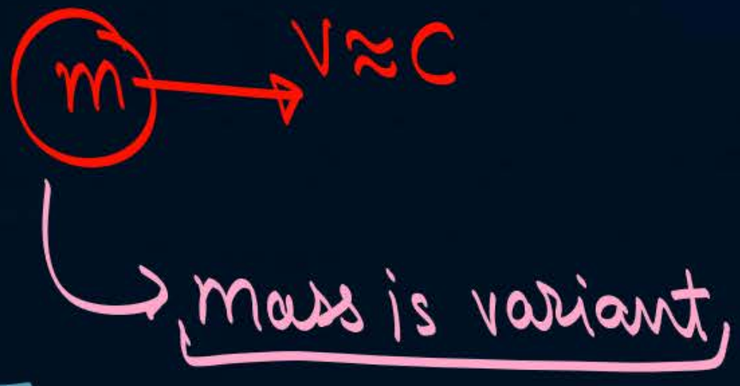
$= \boxed{-1e + x}$

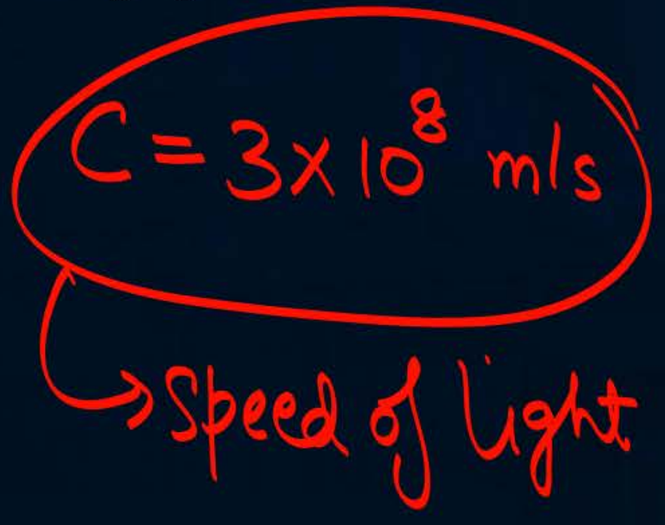
not changing => Constant.

Charge is invariant

changing.

- The value of electric charge on a particle remains the same, no matter its speed or the frame of reference.
- If an electron has a charge of -1.6×10^{-19} C when it's at rest, it will have exactly the same charge even when it's moving very fast (even near the speed of light).
- The charge does not change with motion or location.

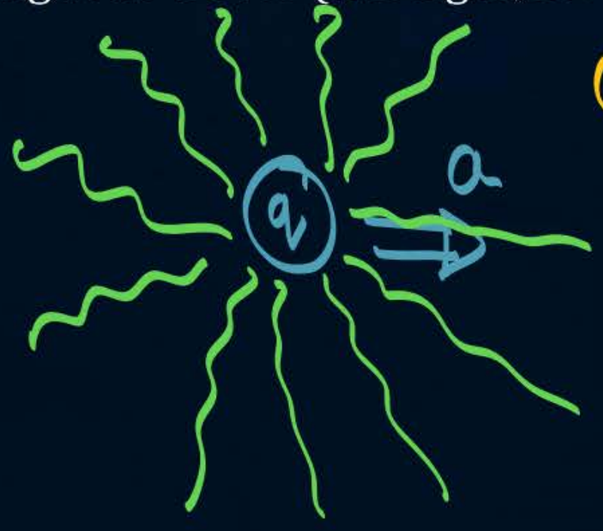




It changes when a velocity $V \approx C$ is given to it.

Accelerated charge radiates energy

- When a charged particle is accelerated (i.e., its velocity changes), it emits electromagnetic radiation and loses energy in the form of waves.
- If a charged particle like an electron is just moving at constant speed, it doesn't radiate.
- But if its speed or direction changes (i.e., it accelerates), it gives off energy as electromagnetic waves (like light, X-rays, etc.).



Charge radiates energy when it is given acceleration.

QUESTION

Which of the following is not a property of electric charge?

A It is Scalar

B It is Vector

C It is Conserved

D It is quantized



HALWA.

$q = ne$

$8e^-$
$5p^+$

QUESTION

If a body has a charge of -4.8×10^{-19} C, how many extra electrons does it have?



A 1

B 2

C 3

D 4

$q = ne$

$-4.8 \times 10^{-19} = (n)(1.6 \times 10^{-19})$

$\Rightarrow n = \frac{-4.8}{1.6} = -3$

$\Rightarrow \boxed{n = -3}$

because of electrons.

QUESTION

How many electrons should be removed from a metal sphere to give it a positive charge of 1C?



$q = 1C$

$e = 1.6 \times 10^{-19} C$

$q = ne$

$\Rightarrow 1C = (n)(1.6 \times 10^{-19} C) \Rightarrow n = \frac{1}{1.6 \times 10^{-19}}$

$\Rightarrow \boxed{n = 6.25 \times 10^{18}}$

Knowledge

So technically, the number of electrons n, in a given charge must be an integer

Now let's refine the calculation:

$$n = \frac{1\,\text{C}}{1.602176634 \times 10^{-19}\,\text{C/electron}} = 6.241509074 \times 10^{18}$$

Rounding to the **nearest integer**:

$$n = \boxed{6{,}241{,}509{,}074{,}460{,}762{,}607} \approx \boxed{6.24 \times 10^{18}}\ \text{electrons}$$

So, the **exact integer number** of electrons in **1 coulomb** is:

$$\boxed{6{,}241{,}509{,}074{,}460{,}762{,}607}\ \text{electrons}$$

Let me know if you'd like a visual or analogy for better understanding!

QUESTION

(CUET 2023)

Which of the following is NOT a property of charges?



A Quantisation

B Scalar

C Additive

D Affected by frames of reference

H.W.

QUESTION

BOARDS PYQ

Calculate the charge carried by 12.5×10^8 electrons?



QUESTION

CHARGE IS INVARIANT

An electron at rest has a charge of 1.6×10^{-19}C. It starts moving with a velocity $v = \frac{c}{2}$, where c is the speed of light, then the new charge on it is:

A 1.6×10^{-19} coulomb

B $1.6 \times 10^{-19}\sqrt{1-\left(\frac{1}{2}\right)^2}$ coulomb

C $1.6 \times 10^{-19}\sqrt{\left(\frac{2}{1}\right)^2-1}$ coulomb

D $\frac{1.6\times10^{-19}}{\sqrt{1-\left(\frac{1}{2}\right)^2}}$ coulomb



QUESTION

Given below are two statements:
Statement-I: A comb run through one's dry hair attracts small bits of papers because the comb gets charged by friction and molecules of the paper gets polarised by the charged comb.
Statement-II: If hair is wet, that comb will not attract the bits of paper.
In the light of the statements, choose the correct answer from the option given below.

A Both Statement-I and Statement-II are true

B Both Statement-I and Statement-II are false

C Statement-I is correct but Statement-II is false

D Statement-I is incorrect but Statement-II is true



DADA → Protons Add या Remove कभी नहीं होंगे।

QUESTION

When a glass rod is rubbed with silk, the glass rod becomes positively charged because:



A. It gains protons

B. It loses electrons

C. It gains electrons

D. Silk gives protons to the glass rod

QUESTION

Which of the following is not a method of charging a body?

A Friction

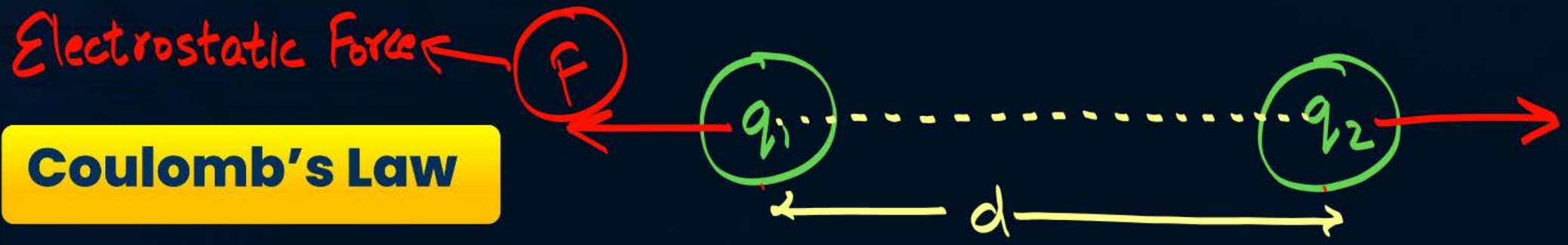
B Conduction

C Induction

D Radiation



Coulomb's Law



- Coulomb's law states that the magnitude of the electrostatic force of attraction or repulsion between two stationary point charges is directly proportional to the product of the magnitudes of charges and inversely proportional to the square of the distance between them.

$$F = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{d^2}$$

(in air or vaccum)

$$\Rightarrow K = \frac{1}{4\pi\epsilon_0}$$

$$F \propto \frac{q_1 q_2}{d^2} \Rightarrow F = K\frac{q_1 q_2}{d^2}$$

Electrostatic constant or Coulomb's constant

$K = 9 \times 10^9\ Nm^2/C^2$

$$K = \frac{1}{4\pi\epsilon_0}$$

Absolute Permittivity of free space (air or vacuum).

$$\epsilon_0 = 8.854 \times 10^{-12}\ C^2 N^{-1} m^{-2}$$

ϵ_m = Absolute permitivity of the medium.

ϵ_r = Relative Permitivity OR Dielectric Constant

$$\Rightarrow \epsilon_r = \frac{\epsilon_m}{\epsilon_0}$$

$5\mu C = 5 \times 10^{-6} C$

$2mm = 2 \times 10^{-3} m$

$K = 9 \times 10^9 \ Nm^2/C^2$

QUESTION

Two point charges of 5 μC and 3 μC are placed at a distance of 2 mm in vacuum. What is the magnitude of the electrostatic force between them?



A 3.375×10^7 N

B 3.375×10^6 N

C 3.375×10^5 N

D 3.375×10^4 N

$$F = \frac{k q_1 q_2}{d^2}$$

$$F = \frac{9 \times 10^9 \times 5 \times 10^{-6} \times 3 \times 10^{-6}}{4 \times 10^{-6}}$$

$$F = \frac{135 \times 10^3}{4} = \frac{13.5}{4} \times 10^4$$

H.W.! Coulomb's laws Ques. practise. → Internet
S.L. Arora.

THANK YOU

